SOLUTIONS

Question 1

(a)

$$A + \bar{A}B = (A + AB) + \bar{A}B = A + B(A + \bar{A}) = A + B$$

[4]

(b)

(i)

$$B + (C \oplus \bar{B})(AB + \bar{C}) = B + (CB + \bar{C}\,\bar{B})(AB + \bar{C}) = B + ABC + \bar{C}\,\bar{B}$$

$$= B + \bar{C}\,\bar{B} = B + \bar{C}$$

[4]

(ii)

$$\overline{AB + AC} + A\,\bar{B}\,C = \overline{AB}\,\overline{AC} + A\,\bar{B}\,C = (\bar{A} + \bar{B})(\bar{A} + \bar{C}) + A\,\bar{B}\,C$$

$$= \bar{A} + \bar{B}\,\bar{C} + A\,\bar{B}\,C = \bar{A} + \bar{B}(\bar{C} + AC) = \bar{A} + \bar{B}(\bar{C} + A) = \bar{A} + \bar{B}\,\bar{C} + A\bar{B}$$

$$= \bar{A} + \bar{B} + \bar{B}\,\bar{C} = \bar{A} + \bar{B}$$

[4]

(c)

$$f = \bar{A}B\bar{C}D + A\bar{B}CD + \bar{A}BC + A\bar{B}C + A\bar{B}\bar{C}D$$

	f	A,B = 00	01	11	10
C,D	00	0	0	0	0
	01	0	1	0	1
	11	0	1	0	1
	10	0	1	0	1

$$f = \bar{A}BD + \bar{A}BC + A\bar{B}D + A\bar{B}C$$
$$= (\bar{A}B + A\bar{B})D + (\bar{A}B + A\bar{B})C$$
$$= (A \oplus B)(C + D)$$

Here, 1 mark for the Karnaugh map, 2 for the correct values and grouping in the map, and 1 for the SOP final expression (first line in Boolean expression above). Remaining 2 marks for converting to a form using one XOR gate (final expression).

[4 + 2]

(d)

Decimal	Hexadecimal	Signed binary (8 bits wide)	Octal
254			375
-17		1110 1111	
	197		627

Give 2 marks per answer.

[6]

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(e)

Z	X	Y	D = next Z
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

Give 2 marks for correct form of truth table, 4 marks for D = next Z

(f)

With two N bit numbers multiplied together, the result has the range:

$$-(2^{2N-2} - 2^{N-1}) \; to \; 2^{2N-2}$$

2 marks for the correct range

This needs 2N bits. Note that it nearly fits into 2N-1 bits, whose range extends to 2^{2N-2} - 1. Give 2 marks for this part.

[4]

g)

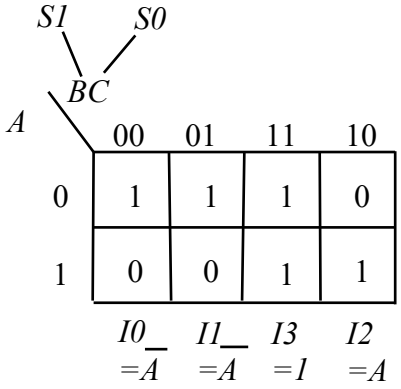
$$f = \bar{A}\,\bar{B} + AB + \bar{A}C$$

This gives the truth table:

A	*B*	*C*	*f*
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

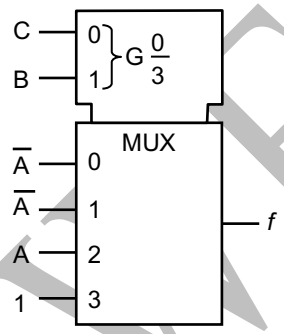
As BC must be used on the select lines of the 4 × 1 MUX, we draw a table with BC arranged along the rows and A along the columns. Furthermore, we choose to connect C = S0 and B = S1.

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2 marks up to this stage

This gives the following implementation:



2 marks for this.

Note: An alternative solution has B = S0 and C = S1. We would then have I0 = A(bar), I1 = A, I3 = 1, and I2 = A(bar).

[4]

(ii) Using two 2 × 1 MUXs, we first need to rearrange the given function in the form of the Boolean function for the output Z for a 2 × 1 MUX, with select input S:

$$Z = \bar{S}I0 + SI1$$

For the given function, we then have:

$$f = \bar{A}\,\bar{B} + AB + \bar{A}C = \bar{A}(\bar{B} + C) + AB$$

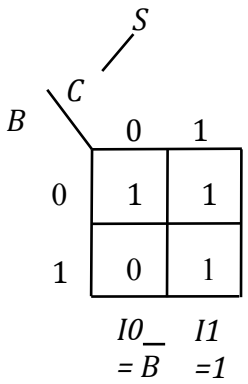
Comparison with the expression for Z gives:

$$I0 = (\bar{B} + C) = g,\ \ I1 = B,\ \ S = A$$

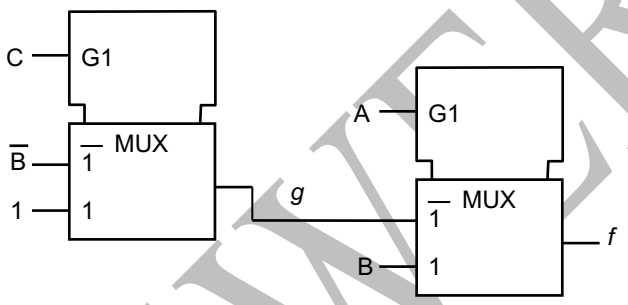
Give 2 marks upto this stage.

We then need to implement $\bar{B} + C$ using a second 2 × 1 MUX. Choosing C for the select line (the alternative of B on the select line is also fine) and inspecting the following map:

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This gives the implementation of the second 2 × 1 MUX, with I0 = B(bar) and I1 = 1. The final implementation is given below (2 marks). Note that other solutions are also possible, full marks should be given for any working solution using the two multiplexers and if needed, some inverters):



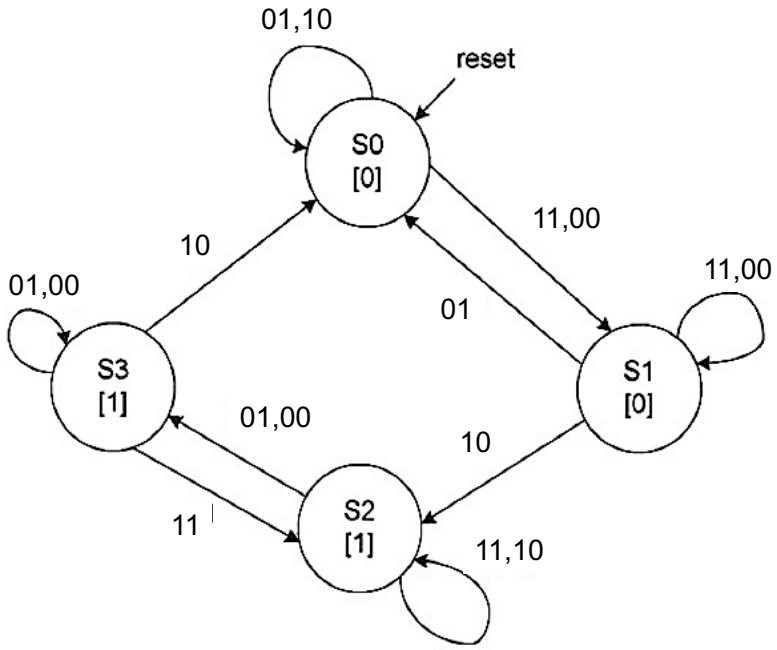
[4]

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Question 2

1. (a) Using the states S0, S1, S2 and S3 to define the FSM, the following state diagram can be obtained. Note that as there are <u>two inputs</u> X1, X2, each state has $2^2 = 4$ transitions possible. Furthermore, a sequence of two input combinations, one after the other, is needed to change the output. For example, for the sequence X1X2 = 11, 10, the first part, X1X2 = 11 causes a transition from S0 to S1, and the second part, X1X2 = 10, causes a transition from S1 to S2, and Z = 0 to Z = 1.

   Give 2 marks for identifying this reasoning.



Give 2 marks for showing four states, 4 for the correct interconnections, and 4 for the correct input/output values.

[12]

(b) Using the encoding system given, the state diagram can be converted into the following table, with Q3Q2Q1Q0 as the present state of the FSM, and $Q3^+Q2^+Q1^+Q0^+$ as the next state:

Q3	Q2	Q1	Q0	X1	X2	$Q3^+$	$Q2^+$	$Q1^+$	$Q0^+$	Z
0	0	0	1	0	0	0	0	1	0	0
0	0	0	1	0	1	0	0	0	1	0
0	0	0	1	1	0	0	0	0	1	0
0	0	0	1	1	1	0	0	1	0	0
0	0	1	0	0	0	0	0	1	0	0
0	0	1	0	0	1	0	1	0	0	0
0	0	1	0	1	0	0	0	0	1	0
0	0	1	0	1	1	0	0	1	0	0
0	1	0	0	0	0	1	0	0	0	1
0	1	0	0	0	1	1	0	0	0	1
0	1	0	0	1	0	0	1	0	0	1
0	1	0	0	1	1	0	1	0	0	1
1	0	0	0	0	0	1	0	0	0	1
1	0	0	0	0	1	1	0	0	0	1
1	0	0	0	1	0	0	0	0	1	1
1	0	0	0	1	1	0	1	0	1	1

Give 0.5 marks for each row.

[8]

(c) As the FSM states are encoded using 1-hot encoding, the Boolean equations can be written directly by inspection:

$$Q0^+ = Q0\overline{X1}X2 + Q0X1\overline{X2} + Q3X1\overline{X2} + Q3X1X2 = Q0\overline{X1}X2 + Q0X1\overline{X2} + Q3X1$$

$$Q1^+ = Q0\overline{X1}\ \overline{X2} + Q0X1X2 + \ Q1\,\overline{X1}\,\overline{X2} + Q1X1X2$$

$$Q2^+ = Q1\overline{X1}X2 + \ Q2\,X1\,\overline{X2} + Q2X1X2 + Q3X1X2 = Q1\overline{X1}X2 + Q2\,X1 + Q3X1X2$$

$$Q3^+ = Q2\overline{X1}\,\overline{X2} + \ Q2\,\overline{X1}\,X2 + Q3\overline{X1}\,\overline{X2} + Q3\overline{X1}\,X2 = Q2\,\overline{X1} + Q3\overline{X1}$$

$$Z = Q3 + Q2$$

Give 2 marks for each Boolean expression.

[10]

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3. Figure 3.2 shows the datapath of a new CPU DECA21 comprising a register file RF1, a binary adder, and other logic. The datapath implements the ALU instructions shown in Figure 3.3. The format of the ALU instructions is shown in Figure 3.1. OP, OPC, d,m,n in Figure 3.2 refer to the unsigned binary values of bit fields OP, OPC, D, M, N in Figure 3.1. Other instructions, not shown here, are executed when OP=1. The W input to RF1 in Figure 3.2 is 1 during every cycle in which the ALU instructions are executed. W may also be 1 when other instructions are executed. Cycles with W=0 do not change any of the datapath registers.

a) For DECA21 state, giving reasons, the number of operands of the Figure 3.3 instructions, the data word size, and number of datapath registers. Discuss whether the architecture is von Neumann, or Harvard, or could be either. [4]

Analysis of a new example. 3 operands: because the three IW fields d,n,m each specifying registers). 8 bit data word size because ALU and register databuses are all 8 bits.The register operands are 6 bits wide =¿ 64 registers. It must be Harvard architecture because the instruction word is different size from the data word. (I will allow an argument that says it could possibly also be von Neumann only if it is stated that this is unlikely).

b) State two distinct types of necessary instructions not shown here that would be expected in any CPU. [2]

bookwork. Jump instructions and memory load/store instructions

c) State the largest positive and negative two's complement values in 8 bits. An 8 bit two's complement subtraction `A := B - C` is defined to *overflow* if and only if the following condition holds:

$$z(\texttt{A(7:0)}) \neq z(\texttt{B(7:0)}) - z(\texttt{C(7:0)})$$

where $z(x)$ is the two's complement interpretation of a binary field x. State the boolean condition on the bits of `A, B, C` that computes overflow. [4]

calculated example and bookwork. +127 -128. We require output sign to be different from that of both of the two inputs: (A7 $\oplus$ B7).(A7 $\oplus$ C7).

d) Using the instructions from Figure 3.3 write in mnemonic form a sequence of instructions that will implement `R0 := 15×R1`. Credit will be given for code that changes the minimum number of registers and uses the minimum number of instructions. [4]

design.
ADD R0, R1, R1
ADD R0,R0, R0
ADD R0, R0, R0
ADD R0, R0, R0
SUB R0, R0, R1
deduct one mark if other registers change. deduct 2 marks if more than 5 instructions.

e) Write in mnemonic form a sequence of two ARM data processing instructions together implementing `R0 := 1023×R1 + 0x0850000`. For each instruction state the precise shift or rotate operation used in the ARM Op2 logic. [4]

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calculated example. RSB R0, R1, R1, LSL #10 logical shift left by 10 - to implement X1024. ADD R0, R0, #0x085000 the op2 number is generated by rotating right 0x85 12 bits.

f) State how input busses `A1,A2,A3` in Figure 3.2 are connected to the instruction word bits `IW(21:0)`. [2]

analysis. A3=D, A1=N, A2=M. IW Bit fields as shown in Figure 3.1.

g) Derive a boolean expression for `FF1.EN`, from `IW(21:0)` and `W`, implementing Figure 3.3. [2]

FF1 is used to save carry (C) from one addition to be used in the next one. FF1.EN = W.not(IW21).not(IW20.not(IW19).not(IW18)). 1 mark deducted if W is not correctly included.

h) Figure 3.4 shows the logic function of block `BX`. Write a truth table implementing the instructions in Figure 3.3 with `OPC` as input and `B1, B0, MUX1.sel, MUX1.0` outputs. Outputs in your truth table which are don't care must be marked `X`. [8]

Design.

OPC	B(1:0)	MUX1.0	MUX1.sel
000	00	0	0
001	01	1	0
010	00	X	1
011	01	X	1
100	10	0	0
101	10	X	1
110	10	1	0
111	11	0	0

Also allow B(1:0)=11, MUX1.0=1, MUX1.sel=0 as alternative for MOV.

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Field	OP	OPC	D	M	N
Bits used	IW21	IW(20:18)	IW(17:12)	IW(11:6)	IW(5:0)

Figure 3.1: Fields in a 22 bit Instruction Word (IW)

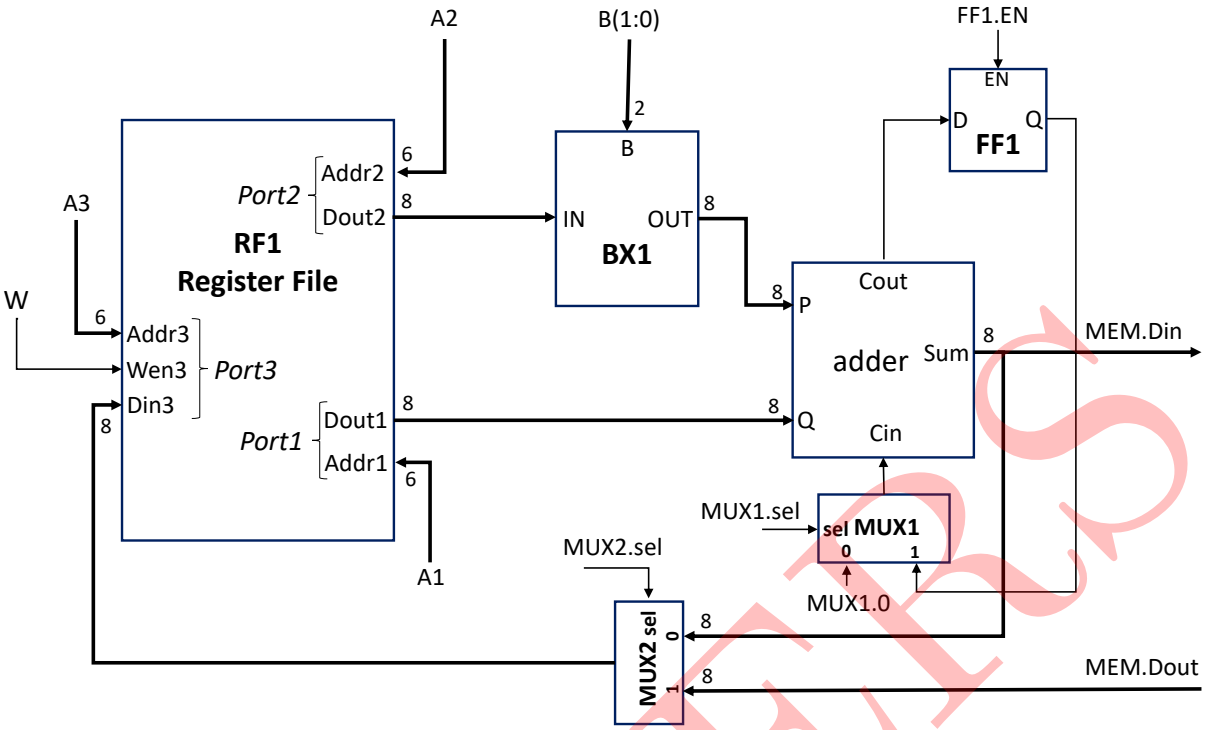


Figure 3.2: Datapath Block Diagram

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Mnemonic Form	OP	OPC	Operation	C
`ADD Rd, Rn, Rm`	0	000	`Rd := Rn + Rm`	write C
`SUB Rd, Rn, Rm`	0	001	`Rd := Rn - Rm`	write C
`ADC Rd, Rn, Rm`	0	010	`Rd := Rn + Rm + C`	write C
`SBC Rd, Rn, Rm`	0	011	`Rd := Rn - Rm + C - 1`	write C
`MOV Rd, Rn`	0	100	`Rd := Rn`	hold C
`MOVC Rd, Rn`	0	101	`Rd := Rn + C`	write C
`MOVP Rd, Rn`	0	110	`Rd := Rn + 1`	write C
`MOVN Rd, Rn`	0	111	`Rd := Rn - 1`	write C

Figure 3.3: DECA21 Instructions: C is written with carry out from the adder where shown, no other instruction writes C

B1	B0	BX.IN	BX.OUT
0	0	*a*	*a*
0	1	*a*	*a* with bits inverted
1	0	X	00000000
1	1	X	11111111

Figure 3.4: Block BX operation

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